

Trigonometry

ANSWER KEY



Right-triangle trigonometric ratios

- 1 In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin \theta$ and the length of the adjacent side.
Adjacent $= \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. $\sin \theta = \frac{5}{13}$.
- 2 In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos \theta$.
Adjacent $= \sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8$. $\cos \theta = \frac{8}{17}$.
- 3 In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.
Opposite $= \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\sin \theta = \frac{24}{25}$, $\tan \theta = \frac{24}{7}$.
- 4 In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin \theta$ and $\cos \theta$ for the angle θ opposite the side of length 20.
 $\sin \theta = \frac{20}{29}$, $\cos \theta = \frac{21}{29}$.

Special angles

- 5 State the exact value of $\sin(30^\circ)$.
 $\sin(30^\circ) = \frac{1}{2}$.
- 6 State the exact value of $\cos(45^\circ)$.
 $\cos(45^\circ) = \frac{\sqrt{2}}{2}$.
- 7 State the exact value of $\tan(60^\circ)$.
 $\tan(60^\circ) = \sqrt{3}$.
- 8 State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.
 $\sin(90^\circ) = 1$ and $\cos(0^\circ) = 1$.

Converting between radians and degrees

- 9 Convert 180° to radians.
 $180^\circ = \pi$ radians.
- 10 Convert 90° to radians.
 $90^\circ = \frac{\pi}{2}$ radians.
- 11 Convert $\frac{5\pi}{6}$ radians to degrees.
 $\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ$.

- 12 Convert 270° to radians.
 $270^\circ = \frac{3\pi}{2}$ radians.
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The Pythagorean identity

- 13 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
 $\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos \theta = \frac{4}{5}$.
- 14 Given $\cos \theta = \frac{5}{13}$ with θ acute, find $\sin \theta$.
 $\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\sin \theta = \frac{12}{13}$.
- 15 Given $\sin \theta = 0.6$, find $\cos^2 \theta$.
 $\cos^2 \theta = 1 - 0.36 = 0.64$.
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Simplifying trigonometric expressions

- 16 Simplify $\sin^2 \theta + \cos^2 \theta + 1$.
 $\sin^2 \theta + \cos^2 \theta = 1$, so the expression equals $1 + 1 = 2$.
- 17 Simplify $\frac{1 - \cos^2 \theta}{\sin \theta}$ (for $\sin \theta \neq 0$).
 $1 - \cos^2 \theta = \sin^2 \theta$, so the expression equals $\frac{\sin^2 \theta}{\sin \theta} = \sin \theta$.
- 18 Simplify $\tan \theta \cos \theta$ (for $\cos \theta \neq 0$).
 $\tan \theta = \frac{\sin \theta}{\cos \theta}$, so $\tan \theta \cos \theta = \sin \theta$.
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Graphs: amplitude and period

- 19 State the amplitude and period of $y = 5\sin(x)$.
Amplitude: 5. Period: 2π .
- 20 State the amplitude and period of $y = 3\cos(2x)$.
Amplitude: 3. Period: $\frac{2\pi}{2} = \pi$.
- 21 State the amplitude and period of $y = 2\sin\left(\frac{x}{3}\right)$.
Amplitude: 2. Period: $\frac{2\pi}{1/3} = 6\pi$.
- 22 A sine graph has amplitude 4 and period π . Write its equation in the form $y = A\sin(Bx)$.
 $A = 4$. $B = \frac{2\pi}{\pi} = 2$. So $y = 4\sin(2x)$.
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Graphs: phase shift and vertical shift

- 23 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{3}$ to the right and 2 units up. Write the new equation.
 $y = \sin\left(x - \frac{\pi}{3}\right) + 2.$
- 24 State the vertical shift and midline of $y = 3\cos(x) + 5.$
Vertical shift: up 5. Midline: $y = 5.$
- 25 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the left. Write the new equation.
 $y = \cos\left(x + \frac{\pi}{4}\right).$
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Solving basic trigonometric equations

- 26 Solve $2\sin\theta = 1$ for $0^\circ \leq \theta < 360^\circ.$
 $\sin\theta = \frac{1}{2}.$ Solutions: $\theta = 30^\circ$ or $\theta = 150^\circ.$
- 27 Solve $\sqrt{2}\cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ.$
 $\cos\theta = \frac{\sqrt{2}}{2}.$ Solutions: $\theta = 45^\circ$ or $\theta = 315^\circ.$
- 28 Solve $\tan\theta = -1$ for $0^\circ \leq \theta < 360^\circ.$
Solutions: $\theta = 135^\circ$ or $\theta = 315^\circ.$
- 29 Solve $2\cos\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ.$
 $\cos\theta = -\frac{1}{2}.$ Solutions: $\theta = 120^\circ$ or $\theta = 240^\circ.$
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Reference angles in other quadrants

- 30 Find $\sin(150^\circ)$ using the reference angle.
The reference angle is 30° ; in Quadrant II, sine is positive. $\sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}.$
- 31 Find $\cos(240^\circ)$ using the reference angle.
The reference angle is 60° ; in Quadrant III, cosine is negative. $\cos(240^\circ) = -\cos(60^\circ) = -\frac{1}{2}.$
- 32 Find $\tan(300^\circ)$ using the reference angle.
The reference angle is 60° ; in Quadrant IV, tangent is negative. $\tan(300^\circ) = -\tan(60^\circ) = -\sqrt{3}.$
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Complementary angle relationships

- 33 Given $\sin(20^\circ) \approx 0.342,$ use the relationship $\sin\theta = \cos(90^\circ - \theta)$ to find $\cos(70^\circ).$
 $\cos(70^\circ) = \sin(20^\circ) \approx 0.342.$

- 34 Given $\sin(35^\circ) \approx 0.574$, find $\cos(55^\circ)$.
 $\cos(55^\circ) = \sin(35^\circ) \approx 0.574$.
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The unit circle: coordinates of key points

- 35 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 60^\circ$.
 $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.
- 36 State the coordinates of the point on the unit circle at $\theta = 180^\circ$.
 $(-1, 0)$.
- 37 State the coordinates of the point on the unit circle at $\theta = 270^\circ$.
 $(0, -1)$.
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Word problems: angles of elevation

- 38 This question has two parts. A drone hovers directly above a landmark. From a point 200 meters from the base of the landmark, the angle of elevation to the drone is 35° .
- a Write an equation relating the drone's height h to the given information, using the tangent ratio.
 $\tan(35^\circ) = \frac{h}{200}$, so $h = 200\tan(35^\circ)$.
- b Find h , to the nearest meter, given $\tan(35^\circ) \approx 0.700$. $h \approx 200(0.700) = 140$ meters.
- 39 This question has two parts. A ladder 10 meters long leans against a wall, making an angle of 65° with the ground.
- a Write an equation for the height h the ladder reaches up the wall, using the sine ratio. $\sin(65^\circ) = \frac{h}{10}$, so $h = 10\sin(65^\circ)$.
- b Find h , to the nearest tenth of a meter, given $\sin(65^\circ) \approx 0.906$. $h \approx 10(0.906) = 9.1$ meters.
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MCQ — Right-triangle trigonometric ratios

- 40 In the right triangle shown, angle θ has opposite side 8 and hypotenuse 17. Find $\sin \theta$.

- a. $\frac{17}{8}$
- b. $\frac{8}{15}$
- c. $\frac{15}{17}$
- d. $\frac{8}{17}$ ✓

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17}.$$

- 41 In a right triangle, angle θ has opposite side 9 and hypotenuse 41. Find $\cos \theta$.

- a. $\frac{9}{41}$
- b. $\frac{9}{40}$
- c. $\frac{41}{40}$
- d. $\frac{40}{41}$ ✓

$$\text{Adjacent} = \sqrt{41^2 - 9^2} = 40; \cos \theta = \frac{40}{41}.$$

- 42 In a right triangle, angle θ has adjacent side 20 and hypotenuse 29. Find $\sin \theta$.

- a. $\frac{21}{20}$
- b. $\frac{20}{29}$
- c. $\frac{21}{29}$ ✓
- d. $\frac{29}{21}$

$$\text{Opposite} = \sqrt{29^2 - 20^2} = 21; \sin \theta = \frac{21}{29}.$$

- 43 In the right triangle shown, the two legs are 9 and 40, and the hypotenuse is 41. Find $\sin \theta$ and $\cos \theta$ (where θ is opposite the side of length 9).

- a. $\sin \theta = \frac{9}{40}, \cos \theta = \frac{40}{9}$
- b. $\sin \theta = \frac{40}{41}, \cos \theta = \frac{9}{41}$
- c. $\sin \theta = \frac{41}{9}, \cos \theta = \frac{41}{40}$
- d. $\sin \theta = \frac{9}{41}, \cos \theta = \frac{40}{41}$ ✓

$$\text{Opposite} = 9, \text{adjacent} = 40, \text{hypotenuse} = 41.$$

44 In a right triangle, angle θ has opposite side 7 and adjacent side 24. Find $\tan \theta$.

a. $\frac{24}{7}$

b. $\frac{24}{25}$

c. $\frac{7}{25}$

d. $\frac{7}{24}$ ✓

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{7}{24}.$$

MCQ — Special angles

45 State the exact value of $\cos(30^\circ)$.

a. 1

b. $\frac{\sqrt{3}}{2}$ ✓

c. $\frac{\sqrt{2}}{2}$

d. $\frac{1}{2}$

$$\cos(30^\circ) = \frac{\sqrt{3}}{2}.$$

46 State the exact value of $\sin(45^\circ)$.

a. 1

b. $\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$ ✓

d. $\frac{1}{2}$

$$\sin(45^\circ) = \frac{\sqrt{2}}{2}.$$

47 State the exact value of $\tan(30^\circ)$.

a. $\frac{\sqrt{3}}{3}$ ✓

b. $\frac{1}{2}$

c. $\frac{\sqrt{3}}{2}$

d. $\sqrt{3}$

$$\tan(30^\circ) = \frac{\sin(30^\circ)}{\cos(30^\circ)} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

48 State the exact values of $\cos(90^\circ)$ and $\sin(0^\circ)$.

a. $\cos(90^\circ) = 0, \sin(0^\circ) = 1$

b. $\cos(90^\circ) = 1, \sin(0^\circ) = 0$

c. Both equal 1

d. Both equal 0 ✓

$$\cos(90^\circ) = 0 \text{ and } \sin(0^\circ) = 0.$$

49 State the exact value of $\tan(45^\circ)$.

a. Undefined

b. $\sqrt{2}$

c. 1 ✓

d. 0

$$\tan(45^\circ) = \frac{\sin(45^\circ)}{\cos(45^\circ)} = 1.$$

MCQ — Converting between radians and degrees

50 Convert 360° to radians.

a. 4π

b. π

c. 2π ✓

d. $\frac{\pi}{2}$

$$360 \times \frac{\pi}{180} = 2\pi.$$

51 Convert 45° to radians.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{4}$ ✓

c. $\frac{\pi}{2}$

d. $\frac{\pi}{8}$

$$45 \times \frac{\pi}{180} = \frac{\pi}{4}.$$

52 Convert $\frac{3\pi}{4}$ radians to degrees.

a. 120°

b. 150°

c. 135° ✓

d. 270°

$$\frac{3\pi}{4} \times \frac{180}{\pi} = 135^\circ.$$

53 Convert 210° to radians.

a. $\frac{7\pi}{12}$

b. $\frac{7\pi}{3}$

c. $\frac{7\pi}{6}$ ✓

d. $\frac{5\pi}{6}$

$$210 \times \frac{\pi}{180} = \frac{7\pi}{6}.$$

MCQ — The Pythagorean identity

54 Given $\sin \theta = \frac{4}{5}$ with θ acute, find $\cos \theta$.

a. $\frac{3}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{5}{3}$

d. $\frac{9}{25}$

$$\cos^2 \theta = 1 - \frac{16}{25} = \frac{9}{25} \Rightarrow \cos \theta = \frac{3}{5}.$$

55 Given $\cos \theta = \frac{7}{25}$ with θ acute, find $\sin \theta$.

a. $\frac{576}{625}$

b. $\frac{24}{25}$ ✓

c. $\frac{25}{24}$

d. $\frac{7}{25}$

$$\sin^2 \theta = 1 - \frac{49}{625} = \frac{576}{625} \Rightarrow \sin \theta = \frac{24}{25}.$$

56 Given $\sin \theta = 0.8$, find $\cos^2 \theta$.

a. 0.2

b. 0.64

c. 0.36 ✓

d. 0.8

$$\cos^2 \theta = 1 - 0.8^2 = 1 - 0.64 = 0.36.$$

57 Given $\cos \theta = \frac{12}{13}$ with θ acute, find $\sin \theta$.

a. $\frac{5}{13}$ ✓

b. $\frac{13}{5}$

c. $\frac{12}{13}$

d. $\frac{25}{169}$

$$\sin^2 \theta = 1 - \frac{144}{169} = \frac{25}{169} \Rightarrow \sin \theta = \frac{5}{13}.$$

MCQ — Simplifying trigonometric expressions

58 Simplify $\sin^2 \theta + \cos^2 \theta - 1$.

a. $2\sin^2 \theta$

b. 0 ✓

c. 2

d. 1

By the Pythagorean identity, $\sin^2 \theta + \cos^2 \theta = 1$, so this is 0 .

59 Simplify $\frac{1 - \sin^2 \theta}{\cos \theta}$ (for $\cos \theta \neq 0$).

a. $\frac{1}{\cos \theta}$

b. $\cos \theta$ ✓

c. $1 - \cos \theta$

d. $\sin \theta$

$$1 - \sin^2 \theta = \cos^2 \theta, \text{ so } \frac{\cos^2 \theta}{\cos \theta} = \cos \theta.$$

60 Simplify $\frac{\sin \theta}{\tan \theta}$ (for $\tan \theta \neq 0$).

a. 1

b. $\sin \theta$

c. $\cos \theta$ ✓

d. $\frac{1}{\cos \theta}$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$, so $\frac{\sin \theta}{\sin \theta / \cos \theta} = \cos \theta$.

61 Simplify $2\sin^2 \theta + 2\cos^2 \theta$.

a. 1

b. 0

c. 2 ✓

d. $4\sin \theta \cos \theta$

$2(\sin^2 \theta + \cos^2 \theta) = 2(1) = 2$.

MCQ — Graphs: amplitude and period

62 State the amplitude and period of $y = 4\sin(x)$.

a. Amplitude 2π , period 4

b. Amplitude 4, period 2π ✓

c. Amplitude 1, period 4

d. Amplitude 4, period π

For $y = A\sin(Bx)$ with $B = 1$: amplitude = 4, period = $\frac{2\pi}{1} = 2\pi$.

63 State the amplitude and period of $y = 2\cos(3x)$.

a. Amplitude 2, period $\frac{2\pi}{3}$ ✓

b. Amplitude 3, period $\frac{2\pi}{2}$

c. Amplitude 6, period 2π

d. Amplitude 2, period 6π

Amplitude = 2; period = $\frac{2\pi}{3}$.

64 State the amplitude and period of $y = 5\sin\left(\frac{x}{2}\right)$.

a. Amplitude 5, period π

b. Amplitude 5, period 4π ✓

c. Amplitude $\frac{5}{2}$, period 2π

d. Amplitude 2, period 4π

$B = \frac{1}{2}$; period = $\frac{2\pi}{1/2} = 4\pi$; amplitude = 5.

65 A cosine graph has amplitude 3 and period 4π . Write its equation in the form $y = A\cos(Bx)$.

a. $y = 3\cos(4\pi x)$

b. $y = 4\pi\cos(3x)$

c. $y = 3\cos\left(\frac{x}{2}\right)$ ✓

d. $y = 3\cos(2x)$

$B = \frac{2\pi}{\text{period}} = \frac{2\pi}{4\pi} = \frac{1}{2}$.

66 The graph shows $y = A\sin(x)$ for some A . Find A by reading the graph's amplitude.

a. $A = 1$

b. $A = 2\pi$

c. $A = 4$

d. $A = 2$ ✓

The curve's peak height (amplitude) is 2.

MCQ — Graphs: phase shift and vertical shift

67 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the right and 3 units up. Write the new equation.

a. $y = \cos(x) + \frac{\pi}{4} + 3$

b. $y = \cos\left(x - \frac{\pi}{4}\right) + 3$ ✓

c. $y = \cos\left(x + \frac{\pi}{4}\right) + 3$

d. $y = \cos\left(x - \frac{\pi}{4}\right) - 3$

Right shift: replace x with $x - \frac{\pi}{4}$; up 3: add 3.

68 State the vertical shift and midline of $y = 2\sin(x) - 4$.

a. Down 4, midline $y = 2$

b. Up 4, midline $y = 4$

c. Down 4, midline $y = -4$ ✓

d. Down 2, midline $y = -4$

Subtracting 4 shifts down 4; the midline is $y = -4$.

69 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{6}$ to the left. Write the new equation.

a. $y = \sin\left(\frac{\pi}{6} - x\right)$

b. $y = \sin(x) + \frac{\pi}{6}$

c. $y = \sin\left(x + \frac{\pi}{6}\right)$ ✓

d. $y = \sin\left(x - \frac{\pi}{6}\right)$

Left shift: replace x with $x + \frac{\pi}{6}$.

70 The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.

- a. $y = 0$
- b. $y = 2$ ✓
- c. $y = 1$
- d. $y = -2$

The curve oscillates evenly about $y = 2$.

MCQ — Solving basic trigonometric equations

71 Solve $2\sin \theta = \sqrt{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $\theta = 45^\circ$ only
- b. $\theta = 45^\circ$ or $\theta = 135^\circ$ ✓
- c. $\theta = 45^\circ$ or $\theta = 225^\circ$
- d. $\theta = 60^\circ$ or $\theta = 120^\circ$

$$\sin \theta = \frac{\sqrt{2}}{2}: \theta = 45^\circ \text{ or } 135^\circ.$$

72 Solve $\sqrt{3}\tan \theta = 3$ for $0^\circ \leq \theta < 360^\circ$.

- a. $\theta = 60^\circ$ or $\theta = 300^\circ$
- b. $\theta = 30^\circ$ or $\theta = 210^\circ$
- c. $\theta = 60^\circ$ or $\theta = 240^\circ$ ✓
- d. $\theta = 60^\circ$ only

$$\tan \theta = \sqrt{3}: \theta = 60^\circ \text{ or } 240^\circ \text{ (tangent positive in Q1 and Q3).}$$

73 Solve $2\cos \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 30^\circ$ or $\theta = 330^\circ$

b. $\theta = 60^\circ$ or $\theta = 240^\circ$

c. $\theta = 60^\circ$ only

d. $\theta = 60^\circ$ or $\theta = 300^\circ$ ✓

$\cos \theta = 0.5$: $\theta = 60^\circ$ or 300° .

74 Solve $\tan \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 45^\circ$ or $\theta = 225^\circ$ ✓

b. $\theta = 45^\circ$ only

c. $\theta = 45^\circ$ or $\theta = 315^\circ$

d. $\theta = 45^\circ$ or $\theta = 135^\circ$

Tangent is positive in Q1 and Q3: $\theta = 45^\circ$ or 225° .

75 Solve $2\sin \theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 210^\circ$ or $\theta = 300^\circ$

b. $\theta = 150^\circ$ or $\theta = 330^\circ$

c. $\theta = 30^\circ$ or $\theta = 150^\circ$

d. $\theta = 210^\circ$ or $\theta = 330^\circ$ ✓

$\sin \theta = -0.5$: $\theta = 210^\circ$ or 330° .

MCQ — Reference angles in other quadrants

76 Find $\sin(120^\circ)$ using the reference angle.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

Reference angle 60° ; sine is positive in Q2: $\sin(120^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}$.

77 Find $\cos(210^\circ)$ using the reference angle.

a. $-\frac{\sqrt{3}}{2}$ ✓

b. $\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

Reference angle 30° ; cosine is negative in Q3: $\cos(210^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$.

78 Find $\tan(330^\circ)$ using the reference angle.

a. $-\frac{\sqrt{3}}{3}$ ✓

b. $\sqrt{3}$

c. $-\sqrt{3}$

d. $\frac{\sqrt{3}}{3}$

Reference angle 30° ; tangent is negative in Q4: $\tan(330^\circ) = -\tan(30^\circ) = -\frac{\sqrt{3}}{3}$.

79 Find $\sin(315^\circ)$ using the reference angle.

a. $-\frac{\sqrt{2}}{2}$ ✓

b. $-\frac{1}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{1}{2}$

Reference angle 45° ; sine is negative in Q4: $\sin(315^\circ) = -\sin(45^\circ) = -\frac{\sqrt{2}}{2}$.

MCQ — Complementary angle relationships

80 Given $\sin(25^\circ) \approx 0.423$, use the relationship $\sin \theta = \cos(90^\circ - \theta)$ to find $\cos(65^\circ)$.

a. -0.423

b. 0.906

c. 0.577

d. 0.423 ✓

$$\cos(65^\circ) = \cos(90^\circ - 25^\circ) = \sin(25^\circ) \approx 0.423.$$

81 Given $\cos(40^\circ) \approx 0.766$, find $\sin(50^\circ)$.

a. 0.643

b. 0.766 ✓

c. 1.306

d. 0.234

$$\sin(50^\circ) = \sin(90^\circ - 40^\circ) = \cos(40^\circ) \approx 0.766.$$

82 Given $\sin(70^\circ) \approx 0.940$, find $\cos(20^\circ)$.

a. 1.064

b. 0.060

c. 0.342

d. 0.940 ✓

$$\cos(20^\circ) = \cos(90^\circ - 70^\circ) = \sin(70^\circ) \approx 0.940.$$

MCQ — The unit circle: coordinates of key points

83 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 30^\circ$.

- a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- b. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

c. $(1, 0)$

d. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

$$\cos(30^\circ) = \frac{\sqrt{3}}{2}, \sin(30^\circ) = \frac{1}{2}.$$

84 State the coordinates of the point on the unit circle at $\theta = 90^\circ$.

a. $(1, 0)$

b. $(0, -1)$

c. $(0, 1)$ ✓

d. $(-1, 0)$

$$\cos(90^\circ) = 0, \sin(90^\circ) = 1.$$

85 State the coordinates of the point on the unit circle at $\theta = 225^\circ$.

a. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

b. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ ✓

c. $(-1, -1)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

$$225^\circ \text{ is in Q3, reference angle } 45^\circ : \text{ both coordinates negative, } -\frac{\sqrt{2}}{2}.$$

86 The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?

a. $\left(\frac{\sqrt{3}}{2}, 0.5\right)$

b. $(0.5, 0.5)$

c. $(1, 0)$

d. $\left(0.5, \frac{\sqrt{3}}{2}\right)$ ✓

$$\cos(60^\circ) = 0.5, \sin(60^\circ) = \frac{\sqrt{3}}{2}.$$

MCQ — Word problems: angles of elevation

- 87 A drone hovers directly above a landmark. From a point 150 meters from the landmark's base, the angle of elevation to the drone is 40° . Find the drone's height, to the nearest meter.

a. 150
b. 196
c. 126 ✓
d. 115

$$h = 150 \tan(40^\circ) \approx 126 \text{ m.}$$

- 88 A ladder leans against a wall, making an angle of 70° with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.

a. 4
b. 12
c. 8
d. 11 ✓

$$h = 12 \sin(70^\circ) \approx 11 \text{ m.}$$

- 89 From the top of an 80 m cliff, the angle of depression to a boat is 25° . Find the horizontal distance from the base of the cliff to the boat, to the nearest meter.

a. 37
b. 80
c. 172 ✓
d. 189

$$\tan(25^\circ) = \frac{80}{d} \Rightarrow d = \frac{80}{\tan(25^\circ)} \approx 172 \text{ m.}$$